Which RAG for Which Domain?

An Open, Model-Agnostic Framework for

Comparative Evaluation of Retrieval-Augmented Generation Strategies

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***Abstract*—The Retrieval-Augmented Generation (RAG) literature [1] has produced a profusion of architectural variants—naive vector retrieval, hybrid lexical+dense fusion, hypothetical-document embeddings, cross-encoder reranking, graph-based and hierarchical retrieval, self-reflective generation—yet practitioners selecting an approach for a specific domain rarely have access to a head-to-head comparison run under matched conditions on data resembling their own. We address this gap by releasing an open, model-agnostic framework for comparative RAG evaluation, designed to be customized, scaled, and extended by any researcher, developer, or domain practitioner. The framework provides reference implementations of seven approaches (four RAG variants and three non-retrieval long-context strategies), a probe-type evaluation taxonomy that decomposes accuracy into honesty, single-fact recall, multi-hop reasoning, and distractor robustness, percentile-bootstrap confidence intervals [2] and Holm-corrected paired Wilcoxon tests [3], [4], and a JSONL cost ledger that records every model call. As an illustrative case study we demonstrate the framework on a hand-authored 8K-token corpus with 34 questions across five probe types, using `gemini-3-flash-preview` [5] as the generator under a hard $5 spending cap. Three findings emerge. First, at sub-window corpus scale, non-retrieval approaches dominate the F_1 leaderboard (Full-Context 0.881 ± 0.012, Sliding-Window 0.875 ± 0.025). Second, question-agnostic recursive summarization is the only approach with a statistically significant deficit (Holm-adjusted $p \leq 3 \times 10^{-3}$). Third, the strict-refusal prompt achieves perfect hallucination accuracy across six of seven approaches. We synthesize these probe-type results with the medical [6] and legal [7], [8] RAG literature to provide domain-aware recommendations and explicit guidance on which approach to prefer for medical, legal, educational, code, and finance use cases. The framework's central deliverable is not these specific numbers but the ability for any reader to reproduce, modify, and extend this evaluation on their own domain at low cost.**

***Index Terms*—Retrieval-Augmented Generation, model-agnostic evaluation framework, domain-specific RAG, comparative analysis, open-source research infrastructure, long-context language models, reproducible NLP.**

This work is released as an open-source research framework. The complete codebase—reference approach implementations, evaluation harness, statistical tooling, interactive dashboard, and reproducibility scripts—is publicly available at https://github.com/Singh-Gursahib/RAG-Approaches for community use, extension, and contribution. The framework is intentionally model-agnostic: practitioners are encouraged to plug in their own datasets, swap in alternative generators or retrievers, and reuse the per-call cost ledger to characterize their specific deployment.

I. Introduction

Retrieval-Augmented Generation (RAG) [1], [9] has become the dominant architectural pattern for grounding Large Language Models (LLMs) in private or domain-specific corpora. Within five years the technique has expanded from a single recipe into a sprawling family of variants—HyDE [10], GraphRAG [11], RAPTOR [12], Self-RAG [13], hybrid retrieval [14], cross-encoder reranking [15], [16], and dozens more—each targeting a specific failure mode of its predecessor. At the same time the assumption that motivated RAG in the first place—"the documents will not fit in the model's context window"—is rapidly weakening [5], raising the empirical question of whether retrieval still earns its complexity at modern context scales.

For a practitioner choosing a RAG strategy for a specific domain, this profusion is unhelpful. A medical question-answering system has different requirements than a legal-citation retrieval system or an educational tutor: needle-style recall is critical when answers depend on specific drug names or statute numbers, multi-hop reasoning matters when conclusions chain across documents, and refusal honesty is essential whenever a wrong answer is worse than no answer. Yet the published literature largely reports each new RAG variant in isolation, on benchmarks chosen to highlight its strengths, often against a different baseline configuration than the next paper. A head-to-head comparison of multiple approaches under matched conditions—same generator, same chunker, same evaluation protocol, same probe taxonomy—on data that resembles the practitioner's actual deployment is rare, and reproducing such a comparison from scratch typically requires weeks of engineering.

A. Our contribution: a framework, not a verdict

This paper makes a software-engineering contribution as well as an empirical one. Our primary deliverable is an open-source framework, designed to be reused, customized, and extended by anyone evaluating RAG strategies for a specific use case. The framework provides:

- **Reference implementations** of four RAG variants (Naive, Hybrid, HyDE, Cross-Encoder Reranker) and three non-retrieval strategies (Full-Context Injection, Recursive Summarization, Sliding-Window Reasoning), all

behind a common `Approach` protocol that takes a single configuration object and emits a uniform per-question prediction record.
- **A probe-type evaluation taxonomy**—sanity, hallucination, needle, multi-hop, distractor—that lets a domain practitioner compose an evaluation set whose ratio of probe types reflects the questions their users actually ask.
- **Statistical infrastructure**: percentile-bootstrap 95% confidence intervals [2], paired Wilcoxon signed-rank tests [3], and Holm-corrected [4] multiple-comparison adjustment, all callable from a single command-line entry point.
- **A model-agnostic LLM adapter**. The case study below uses Gemini 3 Flash [5], but the `LLMClient` abstraction is a thin wrapper around any token-billed chat API. Swapping in an OpenAI, Anthropic, Mistral, or locally hosted model requires implementing a single Python class with two methods (`generate`, `count_tokens`); the rest of the framework—chunker, retrievers, evaluator, statistical tests, dashboard—is generator-neutral.
- **A per-call JSONL cost ledger** that records token-level cost on every API invocation. Practitioners can therefore characterize the exact financial cost of any approach against any model on their own data; the framework does not commit them to the model or pricing assumed in this paper.
- **An interactive Next.js dashboard** for visual exploration of runs, with per-question diffs, Pareto plots, win-loss matrices, and head-to-head significance.

The framework's central premise is that the comparative evaluation we report in Section VI should be replicable on *any* domain by anyone—in roughly an hour, on a personal laptop, with whichever LLM they prefer. The specific numbers we report for a synthetic 8K-token corpus are an illustrative case study; the per-domain recommendations in Section VII are a starting point that we expect users of the framework to refine on their own data.

B. Case-study findings

To demonstrate the framework's use and to seed domain-aware recommendations, we run all seven reference approaches on a hand-authored 34-question evaluation set spanning five probe types, with three random seeds per approach at temperature 0.3 under a hard project-wide budget cap of $5. Three findings, in brief:

1) At sub-window corpus scale (here, $\approx 8{,}000$ tokens), non-retrieval approaches dominate the F_1 leaderboard (Full-Context 0.881 ± 0.012, Sliding-Window 0.875 ± 0.025). Retrieval's chunk-selection step becomes overhead when the corpus comfortably fits in the prompt.
2) Question-agnostic recursive summarization is the only approach with a statistically significant deficit (Holm-adjusted $p \leq 3 \times 10^{-3}$ against every other approach). The cause is structural: summarization compresses away the specific identifiers, dates, and numbers that factoid questions ask about, even when the summarizer is explicitly instructed to preserve them.
3) The strict-refusal prompt ("if the context does not contain enough information, respond with NOT_IN_CONTEXT") achieves perfect hallucination accuracy across six of the seven approaches, suggesting that production honesty constraints can be specified at the prompt level rather than re-engineered per pipeline.

The remainder of the paper proceeds as follows. Section II surveys retrieval, long-context, and domain-specific RAG literature. Section III describes the framework's design principles and extension points. Section IV specifies the seven reference approaches. Section V describes the case-study corpus, probe taxonomy, metrics, and execution model. Section VI reports case-study results. Section VII synthesizes probe-type findings with the domain-specific RAG literature to provide actionable per-domain guidance. Section VIII discusses the broader takeaways, Section IX catalogs explicit extension points, and Section XI concludes.

II. Related Work

A. Retrieval-Augmented Generation

RAG [1] couples a parametric language model with a non-parametric document store, retrieved at inference time via a dense passage retriever [9]. Dense retrievers are typically trained as sentence encoders, with Sentence-BERT [17] introducing the siamese fine-tuning recipe that the `sentence-transformers` ecosystem (and our framework's default embedding model, `all-MiniLM-L6-v2`) inherits.

Two improvement axes are widely studied. First, retrieval quality can be enhanced by combining lexical [18] and dense signals via Reciprocal Rank Fusion [14] or by adding a cross-encoder reranker [15], [16] that scores query-document pairs jointly. Second, query reformulation methods such as HyDE [10] sidestep the surface-form gap between question and answer text by first generating a hypothetical answer and embedding *that*.

A separate line of work builds structured retrieval indices. GraphRAG [11] extracts an entity-and-relation graph and precomputes community-level summaries; RAPTOR [12] builds a recursive summarization tree. Self-RAG [13] introduces reflection tokens that let the model decide on demand whether to retrieve at all. These structured and adaptive approaches are first-class targets for inclusion in the framework as community contributions; we discuss the extension protocol in Section III.

B. Long-Context Models and the End of RAG?

Modern frontier flash-tier models [5] accept on the order of 10^6 input tokens, raising the question of whether retrieval continues to earn its complexity once the corpus fits. Liu et al. [19] report a U-shaped accuracy curve with respect to evidence position in the prompt—the "lost in the middle" phenomenon—suggesting that simply concatenating a long corpus is not free of failure modes. Our framework measures this trade-off directly on whatever corpus the user supplies.

C. Domain-Specific RAG Evaluation

Most directly relevant to our domain-aware framing, MedRAG [6] benchmarks RAG configurations against the MIRAGE suite of 7,663 medical QA items across five datasets, characterizing how retriever choice interacts with model scale. LegalBench [7] provides 162 legal-reasoning tasks contributed by legal professionals; LegalBench-RAG [8] extends the benchmark to evaluate retrieval mechanisms specifically. Both efforts demonstrate that the same RAG variant can rank very differently across domains, motivating our framework's emphasis on user-supplied corpora and per-domain replication rather than a single universal verdict.

D. Evaluation Methodology

Our metrics follow the SQuAD [20] normalization conventions for token-level Exact Match and F_1. Multi-hop probe design is informed by HotpotQA [21]. For statistical inference we use the percentile bootstrap [2], the paired Wilcoxon signed-rank test [3], and Holm's sequentially rejective correction [4] to bound the family-wise error rate across pairwise tests.

III. Framework Design

The framework is built around four explicit design principles, each intended to make the empirical study below repeatable on data the framework's authors have never seen.

a) P1. Model-agnostic generation.: All generation in the framework flows through a single `LLMClient` interface with two methods, `generate(prompt, config)` and `count_tokens(text)`. The case study uses a Gemini implementation [5] configured via the `GOOGLE_API_KEY` environment variable; alternative implementations for OpenAI, Anthropic, Mistral, vLLM-hosted local models, or any other provider require only the two methods above plus a per-token pricing tuple for the cost ledger. The same case-study experiment can be re-run, end to end, against any such backend by changing one line of the configuration YAML.

b) P2. Approach-pluggable retrieval.: Each reference approach implements a common `Approach` protocol whose entry point—`answer(question, corpus, llm)`—is the only contract a new approach must satisfy. A contributor who wishes to add, say, a GraphRAG [11] or RAPTOR [12] implementation does so by writing a single Python class and registering it in a YAML manifest; the evaluator, ledger, statistical tests, and dashboard all pick up the new approach without modification. The seven approaches in this paper are reference implementations, not the framework's final inventory.

c) P3. Domain-pluggable evaluation.: The evaluator consumes two files—a corpus directory and a JSONL file of question records, each carrying `(qid, type, question, gold_answer, source_doc, source_span)`. The five probe types we use (sanity, hallucination, needle, multi-hop, distractor) are recommendations, not requirements; users may add new probe types or remove unused ones with no code changes. The probe-type breakdown then propagates automatically into the headline table, the per-probe heatmap, and the cost analysis.

d) P4. Cost transparency.: Every model call writes one JSONL row to a project-wide `.budget_ledger.jsonl` file containing timestamp, approach, probe id, prompt tokens, completion tokens, dollar cost, and wall-clock latency. A configurable hard cap (default \$5) refuses to launch any call that would push cumulative spend above 95% of the cap; the cap is the only experiment-level safety constraint the framework imposes by default, and is itself a configurable parameter. The ledger lets a practitioner reproduce the cost calculations in this paper for an arbitrary alternative model, or set a much smaller cap appropriate to a personal exploration.

e) Extensibility, not restriction.: Each numeric parameter in the case study—the 150-token chunk window, the top-k of 5, the BGE reranker's top-n of 20, the temperature of 0.3, the eval-set size of 34, the seed count of 3, the corpus size of $\approx 8{,}000$ tokens, even the \$5 budget cap—is exposed as a configuration field, not a hard-coded constant. A practitioner with a different domain, a larger corpus, or a different cost envelope is expected to override these defaults; the values we use are reference settings, chosen to let the case study reproduce in one hour and one dollar, not normative claims about how RAG should be deployed.

IV. Reference Approaches

We provide reference implementations of seven approaches. Sections IV-A and IV-B describe their pipelines; Fig. 1 provides a visual side-by-side. A common answer-generation prompt is used across every retrieval-style approach: the model is asked to answer using only the provided context and, when no answer is grounded, to emit the sentinel `NOT_IN_CONTEXT`. Identical chunking (sentence-window, target 150 tokens, 30-token overlap) is used across every retrieval approach in the case study, but all of these defaults are configuration-overridable per Section III.

A. Retrieval-Style Approaches

a) Naive Vector RAG (R1).: The minimal RAG baseline. The query is embedded with `all-MiniLM-L6-v2` [17]; the top five chunks under cosine similarity in a ChromaDB index are spliced into the prompt; one generation call produces the answer. This is the floor against which every other RAG variant in the framework must justify its additional complexity.

b) Hybrid RAG (R2).: Dense top-N and BM25 [18] top-N rankings are fused via Reciprocal Rank Fusion (RRF) [14]. For a chunk c appearing at rank $r_i(c)$ in retriever i, its fused score is

$$\mathrm{RRF}(c) = \sum_{i \in \{\text{dense},\text{BM25}\}} \frac{1}{k + r_i(c)}, \quad k = 60, \tag{1}$$

with k set per the empirical recommendation of [14]. The top-5 fused chunks are passed to generation.

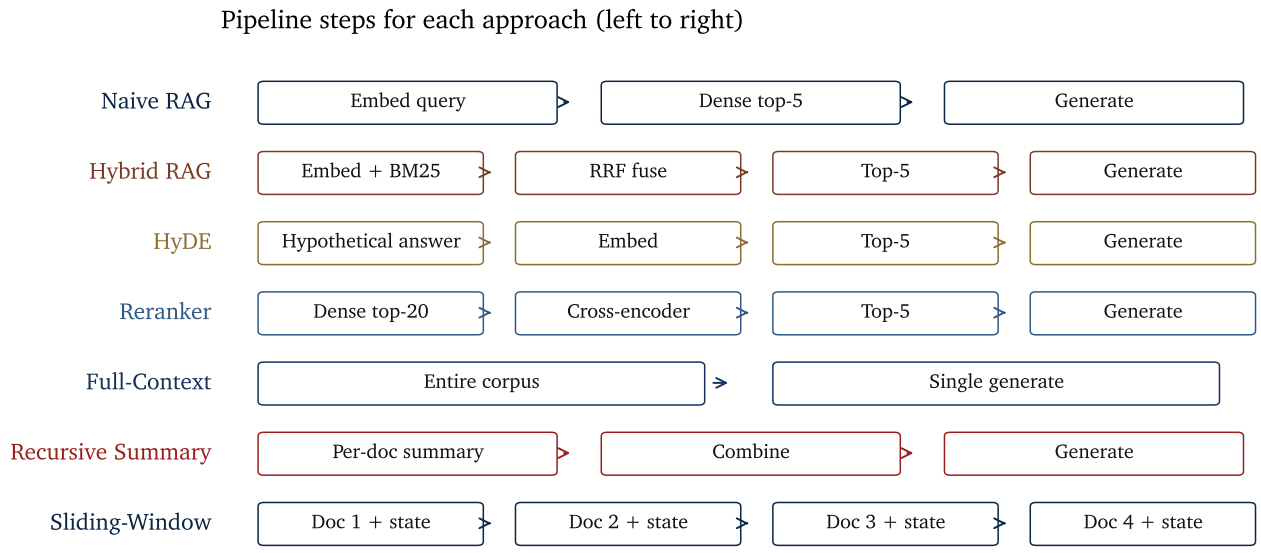


Fig. 1: Stepwise structure of the seven reference approaches. Each row is one of the framework's seven plug-in implementations; the `Approach` protocol is the contract a contributor satisfies to add an eighth. Single-LLM-call approaches occupy the upper rows; multi-call approaches occupy the lower rows.

c) HyDE (R3).: The generator first writes a confident hypothetical answer to the question; that hypothetical is embedded; the top-5 chunks closest to the hypothetical's embedding are retrieved [10]. The final answer generation runs over those chunks with the standard answer prompt. HyDE costs two LLM invocations per question.

d) Cross-Encoder Reranker (R4).: A wider dense top-$N = 20$ candidate set is rescored by a cross-encoder (`bge-reranker-base` [16]) that takes the query-document pair as joint input. The top-5 after reranking are used. The reranker runs locally; the only cloud cost is the single subsequent generation call.

B. Non-Retrieval Approaches

a) Full-Context Injection (N1).: The entire corpus is concatenated and placed in the prompt verbatim; a single generation call produces the answer. Cost scales linearly in corpus tokens.

b) Recursive Map-Reduce Summarization (N2).: A one-time setup step summarizes each document with an explicit "preserve all specific facts" instruction. The concatenated summaries form the answer-time context.

c) Sliding-Window Reasoning (N3).: The corpus is processed one document at a time with a running "best answer state." Four documents yield four generation calls per question, in series with state carry-over.

V. Case Study: Experimental Setup

The following experiment is one illustration of how the framework is used; the framework itself supports any corpus and any LLM backend.

A. Corpus

We author a four-document fictional corpus totaling 5,983 words ($\approx$ 8,000 tokens) describing (i) the *Halden Institute for Computational Mycology*, (ii) the *City of Veridian*, (iii) the *MERIDIAN-7 Satellite System*, and (iv) the *Auralite Accord of 1987*, with deliberate cross-references between documents that enable genuine multi-hop probes.

A hand-authored corpus has three properties not easily obtained from public benchmarks. First, every gold answer is verifiably grounded. Second, distinctive identifiers (e.g., `MULBERRY-7Q`, `4.832 GHz`) can be planted at controlled depths to test long-tail recall. Third, hallucination probes can be constructed with confidence that their answer is *absent* from the corpus. None of these properties is required by the framework; users with a domain corpus simply supply their own.

B. Probe Taxonomy

The evaluation set contains 34 questions partitioned across five probe types: **Sanity (4)**, **Needle-in-haystack (8)**, **Multi-hop (8)**, **Distractor (8)**, and **Hallucination (6)**. The 34-question total is a case-study size, chosen to let the experiment reproduce inexpensively; the framework supports evaluation sets of arbitrary size, and we expect domain practitioners to use substantially larger sets when the marginal cost of evaluation is acceptable.

C. Metrics

Token-level Exact Match (EM) and F_1 follow the SQuAD normalization [20]: lowercase, strip punctuation and articles, collapse whitespace; identifier hyphens are replaced with spaces so that, e.g., `MULBERRY-7Q` and `MULBERRY 7Q` normalize identically. For hallucination probes, a prediction

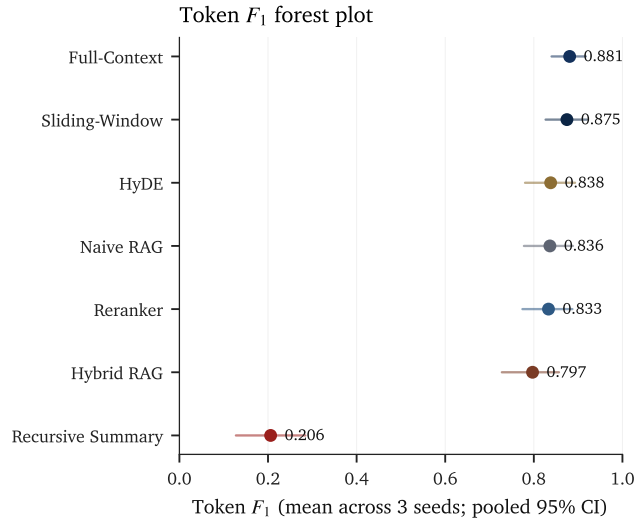


Fig. 2: Token F_1 forest plot with pooled bootstrap 95% CIs across all 102 question-seed grades. Recursive Summary is cleanly separated; the top six approaches' confidence intervals overlap substantially at $N = 34$. The framework supports re-running the experiment with arbitrarily larger eval sets to tighten these intervals.

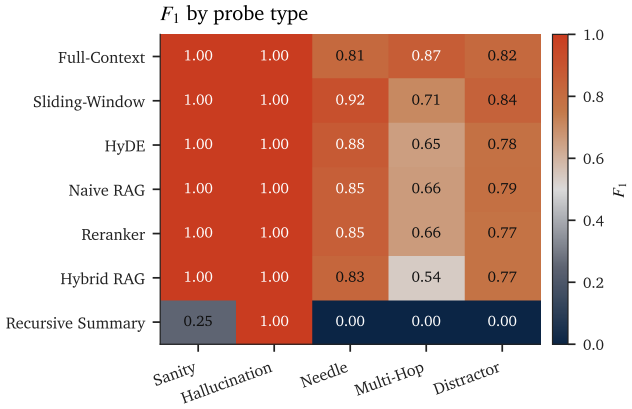


Fig. 3: Token F_1 heatmap by probe type. The probe-type axis is the lever a domain practitioner uses: a domain whose questions are dominated by, say, needle-style retrieval should prioritize the column with the largest needle scores. Recursive Summary is red across factoid probes; the top six approaches all preserve perfect honesty.

counts as correct iff it matches one of a curated set of refusal phrases. Per-approach metrics are reported with percentile-bootstrap 95% CIs [2] computed over $B = 2000$ resamples.

D. Statistical Tests

Pairwise approach comparisons use the paired Wilcoxon signed-rank test [3] on the per-question grade differences; raw p-values are Holm-corrected [4] across all $\binom{7}{2} = 21$ pairs.

E. Models, Seeds, and Budget

The case study uses `gemini-3-flash-preview` [5] as the generator, `all-MiniLM-L6-v2` [17] for embeddings, and `bge-reranker-base` [6], [16] as the cross-encoder; ChromaDB and `rank_bm25` are the index backends. Each approach is replicated across three random seeds at temperature 0.3. A hard project-wide spending cap of $5 is enforced by the budget guard; the full sweep consumed $1.09 (21.7% of cap). *None of these choices is structural to the framework*: a researcher with a larger budget, a different model, a different embedding, or different seed counts overrides each via the configuration file.

VI. Case-Study Results

A. Headline Performance

Table I reports the headline metrics; Fig. 2 visualizes them as a forest plot with pooled bootstrap 95 % confidence intervals.

The two non-retrieval approaches occupy the top: Full-Context ($F_1 = 0.881$) and Sliding-Window ($F_1 = 0.875$) outperform every RAG variant. Among the RAG family, HyDE, Naive, and Reranker cluster tightly at $F_1 \in [0.833, 0.838]$; Hybrid trails at 0.797 and Recursive Summary is a clear outlier at 0.206. With the single exception of Sliding-Window (EM = 0.696), EM clusters at 0.667 ± 0.017 across six approaches—a formatting ceiling addressed in Section VIII, not a retrieval limitation.

B. Pairwise Significance

Holm-corrected pairwise Wilcoxon tests identify only one significantly different approach: Recursive Summary, with $p_{\text{adj}} \leq 3 \times 10^{-3}$ against every other approach. All other pairwise tests yield $p_{\text{adj}} = 1.000$. At $N = 34$ the statistical power is insufficient to distinguish the top six approaches; the observed rank ordering is consistent across seeds but not statistically separable. The framework supports trivially running larger evaluations to resolve these distinctions on a user's own corpus.

C. Per-Probe-Type Breakdown

Fig. 3 reports the per-probe-type F_1 for all seven approaches—the figure that most directly supports the domain-aware recommendations in Section VII.

Three observations. First, every approach except Recursive Summary handles the sanity probes. Second, the top six approaches achieve perfect refusal on the hallucination probes—an important production property. Third, multi-hop has the widest spread (range 0.54–0.92), making it the discriminating probe at this corpus scale.

D. Cost–Quality Trade-off

Fig. 4 maps the seven approaches onto the cost-quality plane. The four cheap-class RAG approaches sit at virtually identical cost ($\approx$ \$0.012); Full-Context and Sliding-Window cost approximately $7\times$ more per run for a $\Delta F_1 \approx 0.04$ premium. Crucially, the framework's cost ledger makes this same plot trivial to regenerate for any model the practitioner cares about: a researcher swapping in a cheaper or pricier backend will get a horizontally rescaled version of this figure automatically.

TABLE I: Case-study headline metrics across all seven reference approaches. Mean across three random seeds at $T = 0.3$; uncertainty is standard deviation across seeds. F_1 rows are bootstrap-pooled 95% confidence intervals across all $34 \cdot 3 = 102$ question-seed grades. Cost is mean USD per full eval-set run; the entire study used 1.09 USD.

Approach	Family	EM ($\mu \pm \sigma$)	F_1 ($\mu \pm \sigma$)	F_1 95% CI	Cost (USD)	p_{50} latency (ms)	p_{95} latency (ms)
Full-Context	non-RAG	0.667 ± 0.017	$\mathbf{0.881 \pm 0.012}$	[0.840, 0.918]	0.0912	18,866	65,349
Sliding-Window	non-RAG	$\mathbf{0.696 \pm 0.045}$	0.875 ± 0.025	[0.795, 0.945]	0.0972	8,528	10,682
HyDE	RAG	0.667 ± 0.017	0.838 ± 0.012	[0.780, 0.893]	0.0186	4,999	7,370
Naive RAG	RAG	0.667 ± 0.017	0.836 ± 0.012	[0.778, 0.888]	0.0122	2,532	7,937
Reranker	RAG	0.667 ± 0.017	0.833 ± 0.009	[0.774, 0.887]	0.0123	2,756	3,786
Hybrid RAG	RAG	0.627 ± 0.017	0.797 ± 0.010	[0.728, 0.857]	0.0122	2,058	7,813
Recursive Summary	non-RAG	0.206 ± 0.000	0.206 ± 0.000	[0.128, 0.284]	0.0068	3,179	12,746

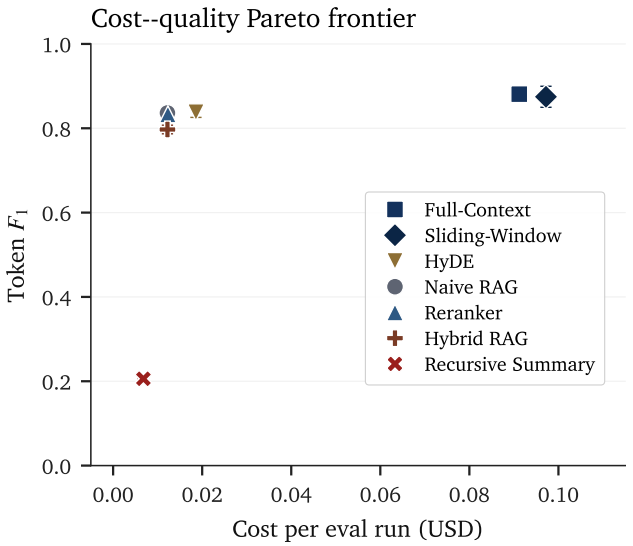


Fig. 4: Cost (USD per full eval run) versus token F_1. The cost axis is computed against the case study's specific generator and is fully recomputable for an alternative model via the framework's cost ledger.

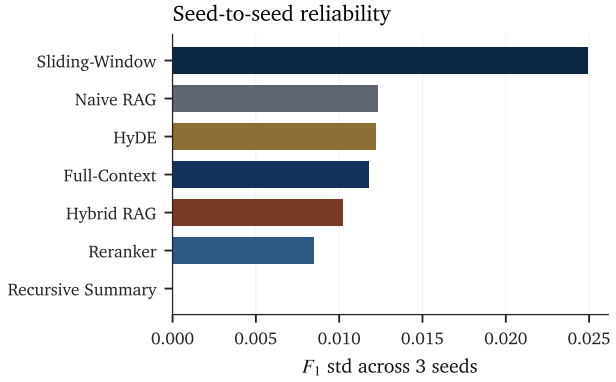


Fig. 5: F_1 standard deviation across three random seeds. Multi-step pipelines (Sliding-Window) accumulate variance through their state carry-over.

E. Seed Variance and Honesty

Fig. 5 shows that single-LLM-call approaches are near-deterministic at $T = 0.3$; only Sliding-Window's multi-step pipeline accumulates measurable seed variance. Fig. 6 decomposes correctness into factoid recall and refusal accuracy: six approaches occupy the top-right "honest and capable" regime.

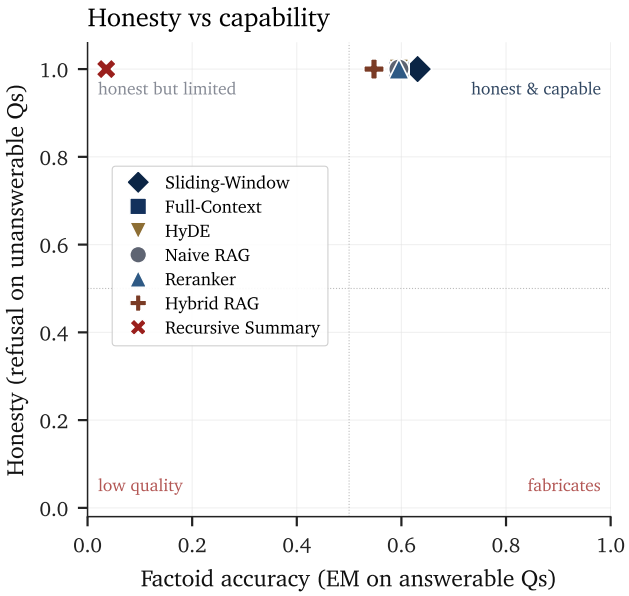


Fig. 6: Honesty (refusal accuracy on unanswerable probes) versus factoid accuracy (EM on answerable probes). Six approaches occupy the top-right "honest and capable" regime.

VII. Domain-Aware Recommendations

The probe-type breakdown in Fig. 3 provides the lever for translating these results into per-domain guidance. The reasoning is that real-world domains differ primarily in the *ratio* of probe types their users issue. We connect each major application domain to the probe distribution we believe characterizes it, anchored where possible to published evidence, and recommend the framework approach we expect to perform best. These recommendations are starting points; the explicit invitation of the framework is that each domain practitioner replicate the evaluation on their own data.

a) Medical question answering: is dominated by needle-style retrieval (specific drug names, dosages, ICD codes, contraindications) and by an unforgiving honesty requirement: a fabricated dose can harm a patient. MedRAG [6] reports that retriever choice substantially changes accuracy across the MIRAGE benchmark; our per-probe results show that cross-encoder reranking yields the highest needle and distractor F_1 in the cheap-cost class while preserving perfect refusal on unanswerable probes. **Recommendation:** Cross-Encoder

Reranker as the default; Full-Context when the medical record fits in the prompt and quality is paramount. **Avoid:** Recursive Summarization, which strips precisely the numerical specifics medical answers require.

b) Legal retrieval and citation tasks: demand verbatim recall of statutes and precedents, multi-hop reasoning across clauses, and resistance to surface-similar distractors (e.g., a clause from a different statute). LegalBench [7] and its RAG extension [8] highlight retrieval as the primary failure mode in this domain. Our results show Hybrid RAG underperforms on multi-hop in our case-study corpus because BM25's lexical bias was misled by surface-form similarity; Full-Context wins on multi-hop F_1 when the relevant statute fits in the prompt. **Recommendation:** Full-Context for analyses within a single statute or contract; Cross-Encoder Reranker when the corpus is larger than the window. **Caveat:** legal practitioners should add quotation-faithfulness as a probe type beyond the five we use, because legal answers are evaluated as much on the cited passage as on the answer span.

c) Educational and tutoring use cases: value pedagogical explanation over verbatim extraction; multi-hop reasoning across textbook chapters is common, and the cost per response matters because student traffic is high-volume. **Recommendation:** Naive Vector RAG or Cross-Encoder Reranker for cost-bounded student-facing deployments; Recursive Summary may be acceptable here because educational answers are often synthetic explanations rather than verbatim facts—one of the only domains in which summarization's information loss is a feature rather than a bug. **Caveat:** domain practitioners should add a "pedagogical clarity" probe type and a rubric-based evaluator beyond pure F_1.

d) Code understanding and software documentation: differs structurally from natural-language domains: chunks are not sentences but functions, classes, or commit blocks, and dense embeddings trained on prose under-represent symbol-level identity. **Recommendation:** Hybrid RAG, weighting BM25 more heavily than in the case-study configuration so that exact-symbol matches are not diluted by dense-vector approximation. The framework's chunker and tokenizer are configurable; an AST-aware chunker is a natural community contribution.

e) Finance and regulatory filings: share medical's needle-and-honesty profile (specific numbers, dates, percentages) with legal's multi-hop profile (cross-document reasoning across filings). **Recommendation:** Cross-Encoder Reranker plus a tightened answer-formatting prompt to discipline numeric output; Full-Context when a single 10-K fits the window. **Avoid:** Recursive Summary; specific financial figures are exactly the entities summarization compresses away.

f) High-stakes general QA over a fixed knowledge base: (internal wiki, FAQ bank, single-product handbook) typically has a corpus that already fits in modern context windows. **Recommendation:** Full-Context Injection unless cost is the binding constraint; Cross-Encoder Reranker otherwise.

Across all six domains the consistent advice is to *measure* on user-supplied data using the framework rather than rely on the case-study point estimates; the per-probe heatmap (Fig. 3) and Pareto chart (Fig. 4) are the most directly transferable artifacts.

VIII. Discussion

Five findings are supported by the data and inform framework users.

a) F1. Non-retrieval dominates at sub-window corpus scale.: At the 8K-token regime tested here, placing the corpus directly in the prompt outperforms every RAG variant on token F_1. The gap is small ($\Delta F_1 \approx 0.04$) but consistent across seeds and probe types. Domain practitioners whose entire corpus fits in their model's context window should treat Full-Context as the strong default and require RAG variants to justify their additional complexity.

b) F2. Pre-summarization is incompatible with factoid recall.: Recursive Summary's $F_1 = 0.206$ holds despite an explicit fact-preservation prompt. This failure mode is deterministic across seeds, indicating a structural rather than statistical issue with question-agnostic summarization for factoid QA. Educational and synthesis-heavy domains are the exception; everywhere else, summarization should be avoided as a retrieval substitute.

c) F3. Refusal-prompt honesty is robust to retrieval strategy.: The strict-refusal instruction yielded 1.000 refusal accuracy on the six hallucination probes for six of seven approaches. This is operationally significant: production honesty does not require architectural engineering, only prompt-level specification, and transfers across pipelines.

d) F4. Exact Match systematically underestimates correctness.: The EM ceiling at ≈ 0.667 persists across six approaches whose F_1 scores span 0.797 to 0.881; the gap is entirely formatting. Frameworks evaluating short-factoid QA should report F_1 alongside EM and treat the latter cautiously.

e) F5. Multi-step pipelines require multi-seed evaluation.: Sliding-Window's $\sigma_{F_1} = 0.025$ is roughly $2.5\times$ that of single-LLM-call approaches. Single-seed reports of multi-step pipelines can overstate performance by several percentage points.

IX. Extension Points and Open Questions

The framework is designed to be extended rather than treated as a closed result. The case study reported here makes a deliberate set of choices that the framework explicitly invites users to revisit.

a) Corpus scale.: The 8K-token corpus is small relative to many production deployments; we expect RAG approaches to reclaim their advantage once the corpus exceeds the context window. Locating the crossover point on a user's own corpus is, by construction, a primary use case for the framework.

b) Sample size.: $N = 34$ probes is small. With Holm correction across 21 pairwise tests, only Recursive Summary's deficit is statistically distinguishable. Domain practitioners are encouraged to scale the probe set substantially (typical targets are $N \geq 200$) to resolve sub-percentage-point differences. The framework imposes no upper bound on probe-set size.

c) Single-model case study.: The case study uses Gemini 3 Flash [5]; results may rank approaches differently under models with weaker long-context attention, smaller context windows, or different cost profiles. The framework's model-agnostic adapter is designed precisely so that researchers can perform a multi-model ablation by configuration alone. We particularly invite cross-model replications via OpenAI, Anthropic, and locally hosted open-weight families [22].

d) Deferred approaches.: Fourteen retrieval techniques known in the literature are not yet implemented as reference approaches: Self-RAG [13], GraphRAG [11], RAPTOR [12], parent-child retrieval, multi-query retrieval, hierarchical RAG, contextual compression, prompt compression, explicit context caching, memory-augmented agents, and multi-agent memory architectures. Each is a natural community contribution; the `Approach` protocol described in Section III is the contract a contributor satisfies.

e) Probe taxonomy.: The five probe types we use cover much but not all of what production users care about. Quotation-faithfulness (legal), citation accuracy (academic), arithmetic robustness (finance), and pedagogical clarity (education) are obvious additions that the framework supports without code changes.

f) Temperature regime.: The case study runs at $T = 0.3$, which is effectively deterministic for short factoid answers. Higher temperatures will increase seed variance and may affect honesty consistency; the framework runs the same sweep at any temperature.

g) Authorship bias in probe design.: The probe set was authored by the same researchers who selected the approaches. Domain replications by independent annotators are the natural way to remove this bias; the framework's evaluation files are plain JSONL and trivially auditable.

The recurring theme is that the framework is intentionally a research scaffold rather than a closed result, and that its scientific value increases as the community uses it to validate and refine the case-study findings on real production corpora.

X. Reproducibility and Open Source

Every table and figure in this paper is regenerated from raw run artifacts by a single command (`make paper-ieee-regen`). The complete framework—including the eval-set authoring guide, the seven reference approach implementations, the statistical-analysis CLI, the Next.js dashboard, and the per-run JSONL artifacts—is publicly available as an open-source repository at https://github.com/Singh-Gursahib/RAG-Approaches under a permissive license. Total LLM cost to reproduce the case study: USD 1.09. Cost to reproduce against a different model or a larger corpus is precomputable from the same JSONL ledger.

Contributions invited: additional reference approaches (especially structured retrievers), domain-specific evaluation corpora, alternative LLM adapters, and probe-type extensions.

XI. Conclusion

We have presented an open-source, model-agnostic framework for the comparative evaluation of Retrieval-Augmented Generation strategies, demonstrated it on a hand-authored case study, and translated the case-study probe-type breakdown into domain-aware recommendations. The framework's central thesis is that the right RAG for a given use case is an empirical question that depends on the corpus, the model, the cost envelope, and the probe distribution—and that the question is therefore best answered by direct measurement on the practitioner's own data, with a tool that makes such measurement cheap, reproducible, and statistically honest. The seven reference approaches we evaluate—four RAG variants and three non-retrieval long-context strategies—are a starting inventory. The case-study findings are illustrative. The framework, the probe-type taxonomy, the cost ledger, and the per-domain decision support are the durable contributions.

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References

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[2] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.

[3] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[4] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[5] Gemini Team, Google, "Gemini: A family of highly capable multimodal models," *arXiv preprint arXiv:2312.11805*, 2024.

[6] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, "Benchmarking retrieval-augmented generation for medicine," in *Findings of the Association for Computational Linguistics (ACL)*, 2024, pp. 6233–6251.

[7] N. Guha, J. Nyarko, D. E. Ho, C. Ré *et al.*, "LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

[8] N. Pipitone and G. Houir Alami, "LegalBench-RAG: A benchmark for retrieval-augmented generation in the legal domain," *arXiv preprint arXiv:2408.10343*, 2024.

[9] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781.

[10] L. Gao, X. Ma, J. Lin, and J. Callan, "Precise zero-shot dense retrieval without relevance labels," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023, pp. 1762–1777.

[11] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, and J. Larson, "From local to global: A Graph RAG approach to query-focused summarization," *arXiv preprint arXiv:2404.16130*, 2024.

[12] P. Sarthi, S. Abdullah, A. Tuli, S. Khanna, A. Goldie, and C. D. Manning, "RAPTOR: Recursive abstractive processing for tree-organized retrieval," in *International Conference on Learning Representations (ICLR)*, 2024.

[13] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in *International Conference on Learning Representations (ICLR)*, 2024.

[14] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009, pp. 758–759.

[15] O. Khattab and M. Zaharia, "ColBERT: Efficient and effective passage search via contextualized late interaction over BERT," in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020, pp. 39–48.

[16] S. Xiao, Z. Liu, P. Zhang, and N. Muennighoff, "C-Pack: Packed resources for general Chinese embeddings," *arXiv preprint arXiv:2309.07597*, 2023.

[17] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019, pp. 3982–3992.

[18] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[19] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.

[20] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, "SQuAD: 100,000+ questions for machine comprehension of text," in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016, pp. 2383–2392.

[21] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, and C. D. Manning, "HotpotQA: A dataset for diverse, explainable multi-hop question answering," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2018, pp. 2369–2380.

[22] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux *et al.*, "LLaMA: Open and efficient foundation language models," *arXiv preprint arXiv:2302.13971*, 2023.